

Overlap-Corrected Full-Block Cover Certificates: Incidence Geometry, Dual Obstructions, and the TPC-18 Interface

Liang Wang

School of Artificial Intelligence and Automation
Huazhong University of Science and Technology, Wuhan 430074, P.R. China
wangliang.f@gmail.com

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Abstract

The all-block cover hypothesis in TPC-18 is a physical asymptotic representation with a controlled $o(X)$ remainder, not a consequence of bounded overlap. We formulate a stronger exact sufficient certificate on the literal finite atom space. The columns of a weighted incidence matrix B are the exact candidate block coefficients, and the zero-remainder route is the constrained synthesis equation $Bc = w$, where w is the physical target coefficient. For unrestricted coefficients, the orthogonal residual $(I - BB^\dagger)w$ is the unique least-squares residual and, when nonzero, yields a sharp obstruction through an exact dual variational formula. For nonnegative coefficients, the corresponding cone projection gives both a quantitative residual and a Farkas certificate. More general convex coefficient rules whose attainable images are closed have an analogous support-function dual.

Two explicit families show why overlap bookkeeping alone is insufficient. Even a partition of all atoms into disjoint two-point blocks can stay a fixed relative distance $1/\sqrt{10}$ from the target, and a two-column system may admit a signed exact cover while having nonnegative distance $1/\sqrt{2}$. We also prove that a cover after collapsing native keys need not lift to a literal cover. Least-squares conditioning, residual stability, and the exact coefficient-amplification cost are recorded. Most importantly, only an exact literal identity transports the original global normalization without an additional remainder theorem.

Applied to TPC-18, the framework is an L1 certificate schema for the exact zero-remainder route through the still-unproved all-block representation gate. It neither constructs that exact certificate nor controls a nonzero physical remainder, and it does not prove the primitive determinant-dispersion estimate. The determinant/zero-mode condition $\lambda_D \leq 2\eta_Z$, the strict physical-loss rule $\Lambda_{\text{phys}} < 1/400$, and the TPC-18 all-block representation gate remain independent. No L2 fixed-shift arithmetic estimate, parity breakthrough, or prime-pair conclusion is claimed.

Keywords: exact block cover; incidence matrix; bounded overlap; Farkas lemma; least-squares residual; fixed shift; determinant dispersion.

Literal-frame convention. Every interface statement concerns one prescribed nonzero shift h_0 , the actual active atoms of the physical packet, their native keys, and the original global normalization. Equal determinant values are not used to identify distinct atoms. No statement is specialized to $h_0 = 2$.

1 Introduction

The TPC-18 stopping argument isolates a primitive large-determinant/low-gcd residual inside each admissible symmetric tail block [5]. Its conditional global conclusion requires more than a good estimate on each block: the physical hard packet must be represented by the entire proposed block family, with the correct unit coefficients, overlaps, masks, and normalization, up to a controlled $o(X)$ physical remainder. That requirement was deliberately left as an all-block representation hypothesis.

A bounded-overlap partition is often enough for inequalities because one may estimate each atom a bounded number of times. It is not enough for an identity. If two blocks overlap, the shared atom

receives the sum of both block coefficients. If the block weights cannot reproduce the target coefficient exactly, dividing by a support multiplicity does not repair signs, phases, masks, or nonconstant block profiles. This distinction is especially important when a later argument is calibrated to a single global main term. An exact coefficient identity is a sufficient, zero-remainder way to verify this hypothesis. It is stronger than the original asymptotic statement: a nonzero coefficient residual may also be admissible if a separate theorem proves that its physical image is $o(X)$.

The purpose of this paper is to turn that qualitative warning into an exact finite-dimensional certificate. Let Ω_X be the actual atom set and let w_X be the literal coefficient of the physical packet. Each candidate block contributes one exact column b_j . The synthesis matrix

$$B_X = (b_1 \cdots b_m)$$

contains all overlap information. A zero-remainder cover is precisely

$$B_X c_X = w_X,$$

together with the coefficient restrictions demanded by the decomposition. Linear algebra then separates three questions:

- (i) does the target lie in the column space?
- (ii) if signed coefficients are forbidden, does it lie in the allowed cone or polytope?
- (iii) if it does, what coefficient amplification is paid when the block estimates are reassembled?

1.1 Main results

The first result is the exact subspace certificate. With $P_B = BB^\dagger$,

$$\inf_c \|w - Bc\|_2 = \|(I - P_B)w\|_2.$$

The residual is zero exactly when a cover exists. When it is nonzero, its normalized direction annihilates every candidate block while detecting the target, so it is a short dual certificate of failure.

The second result is the constrained analogue. For real nonnegative coefficients, a cover exists exactly when

$$B^\top y \geq 0 \implies \langle y, w \rangle \geq 0.$$

Failure therefore has a Farkas witness. Cone projection yields the quantitative distance formula

$$\text{dist}(w, B\mathbb{R}_+^m) = \max_{\substack{\|y\|_2 \leq 1 \\ B^\top y \leq 0}} \langle y, w \rangle.$$

A support-function formula handles any closed convex allowed coefficient image.

The third result consists of sharp counterexamples. A disjoint partition of $2n$ atoms into n two-point blocks has overlap one, covers every atom as a set, and yet may have relative least-squares defect exactly $1/\sqrt{10}$ for every n . A separate two-column example has an exact signed cover but no nonnegative cover, with cone distance $1/\sqrt{2}$. Thus neither bounded overlap nor full set coverage implies exact or asymptotically approximate coefficient coverage.

Finally, we state the only automatic normalization transfer. If $Bc = w$ exactly, then every linear physical statistic with its original normalizer satisfies the exact reassembly identity, and the block error is amplified by $\|c\|_1$. If $w = Bc + r$ only approximately, the physical remainder is the additional term carried by r . Small Euclidean residual alone does not remove it, although a separate physical bound on that term can verify the original asymptotic TPC-18 route.

1.2 Claim levels

The projection identities, dual formulas, counterexamples, and conditioning bounds are unconditional L0 statements. Their literal placement on the TPC-18 atom space gives an exact L1 certificate schema. We do not construct the physical incidence matrix from the whole preceding normal form, prove its feasibility, or prove the primitive determinant-dispersion estimate. Consequently there is no new L2 arithmetic theorem.

1.3 Organization

Section 2 defines the literal atom and block spaces. Section 3 gives exact subspace, cone, and convex certificates. Section 4 proves the overlap and coarsening obstructions. Section 5 records conditioning and normalization costs. Section 6 states the precise conditional crosswalk to TPC-18. Sections 7 and 8 keep the remaining gates and claim boundary explicit.

2 The literal atom–block model

2.1 Atoms, native keys, and weighted incidence

Fix X and one prescribed nonzero shift h_0 . Let $\Omega = \Omega_X$ be a finite set of *literal atoms*. An atom retains every key needed to reconstruct its physical contribution: source variables, divisor variables, dyadic masks, determinant row, residue data, and any provenance label. Two atoms with the same numerical determinant remain distinct when their other native keys differ.

Let $\mathbb{K}$ be either $\mathbb{R}$ or $\mathbb{C}$, and equip $\mathbb{K}^\Omega$ with the Euclidean inner product and norm. The target coefficient is

$$w = (w_\omega)_{\omega \in \Omega} \in \mathbb{K}^\Omega.$$

The set Ω is the actual active universe: it contains both the support of w and every atom emitted by a candidate block. Thus a column that produces an unwanted atom is represented by a nonzero entry in a row where $w_\omega = 0$; such leakage cannot be deleted by restricting the matrix after the fact.

Let $\mathcal{J} = \{1, \dots, m\}$ index the candidate blocks. For block j , let

$$I_{\omega j} = \mathbf{1}_{\{\omega \text{ occurs in block } j\}}$$

be its support incidence and let $g_{\omega j} \in \mathbb{K}$ be its exact unit block coefficient. The *weighted incidence matrix* is

$$B_{\omega j} = I_{\omega j} g_{\omega j}. \quad (1)$$

Its j -th column b_j is therefore the literal coefficient vector delivered by one unit of that block. In the unweighted set cover model $g_{\omega j} = 1$, but the weighted form is essential when masks, signs, or phases vary inside a block.

Definition 2.1 (Overlap degree). The support overlap degree is

$$\Delta(I) = \max_{\omega \in \Omega} \sum_{j=1}^m I_{\omega j}.$$

It counts how many candidate columns touch an atom. It contains no information about the values $g_{\omega j}$, the target w_ω , or the existence of a synthesis coefficient.

2.2 Exact and constrained covers

Definition 2.2 (Literal cover). Let $\emptyset \neq \mathcal{C} \subseteq \mathbb{K}^m$ be the allowed coefficient set. A vector $c \in \mathcal{C}$ is an *exact literal cover* when

$$Bc = w. \quad (2)$$

The cover defect is

$$\varepsilon_{\text{cov}}(B, w; \mathcal{C}) = \text{dist}(w, B\mathcal{C}) = \inf_{c \in \mathcal{C}} \|w - Bc\|_2. \quad (3)$$

Remark 2.3 (Membership versus zero distance). Exact feasibility is the membership statement

$$w \in B\mathcal{C}.$$

If $B\mathcal{C}$ is closed, this is equivalent to $\varepsilon_{\text{cov}} = 0$. Without closedness, zero distance need not be attained and is not an exact cover. Every later use of $\varepsilon_{\text{cov}} = 0$ as a binary certificate is therefore accompanied by closedness; otherwise the membership statement is used directly.

The relevant choices of $\mathcal{C}$ include:

$$\mathbb{K}^m \quad (\text{unrestricted signed or complex synthesis}), \quad \mathbb{R}_+^m \quad (\text{nonnegative synthesis}),$$

as well as affine or polyhedral rules that encode a prescribed partition of unity. The coefficient rule is part of the theorem: an unrestricted solution does not certify a nonnegative cover.

Remark 2.4 (No multiplicity division). Replacing every target coefficient by $w_\omega / \sum_j I_{\omega j}$ is not an overlap correction. It changes the target and ignores the block profiles $g_{\omega j}$. The only exact correction in the present model is solving (2).

2.3 Determinant compatibility is metadata, not coverage

For every atom ω , let $\mathfrak{d}(\omega)$ denote its literal determinant key and let $\mathfrak{p}(\omega)$ denote the rest of its provenance. A candidate column is called *TPC-18 compatible* only if every nonzero entry:

- (a) belongs to the same prescribed shift h_0 ;
- (b) obeys the required block geometry and masks;
- (c) carries its native pair $(\mathfrak{d}(\omega), \mathfrak{p}(\omega))$; and
- (d) uses the same physical atomic normalization as w .

Compatibility is checked before B is formed. A solution using an incompatible column is not a physical cover.

Proposition 2.5 (Compatibility and coverage are independent). *Neither of the following implications is valid in general:*

- (i) *every column is determinant-compatible $\implies w \in \text{ran } B$;*
- (ii) *$w \in \text{ran } B \implies$ the columns are determinant-compatible.*

Proof. For (i), take any compatible nonzero column b_1 and a target not in its one-dimensional span. For (ii), append to a valid synthesis system an algebraically identical column whose provenance metadata is wrong; its span is unchanged. Column-space membership alone cannot inspect metadata. $\square$

Remark 2.6 (Determinant energy). The Gram matrix $G = B^*B$ records column overlap and, under an exact cover, gives $\|w\|_2^2 = c^*Gc$. This identity is not a lower bound for the literal determinant energy in another coefficient space and does not estimate a distinguished zero mode. Such arithmetic comparisons remain separate inputs.

3 Projection and dual certificates

3.1 Unrestricted coefficients

Let $B^\dagger$ be the Moore–Penrose pseudoinverse and let

$$P_B = BB^\dagger$$

be the orthogonal projection onto $\text{ran } B$ [2]. Write

$$q_* = P_B w, \quad r_* = (I - P_B)w. \tag{4}$$

Theorem 3.1 (Exact subspace cover certificate). *For $B \in \mathbb{K}^{n \times m}$ and $w \in \mathbb{K}^n$,*

$$\inf_{c \in \mathbb{K}^m} \|w - Bc\|_2 = \|r_*\|_2, \tag{5}$$

$$\|r_*\|_2 = \max_{\substack{y \in \mathbb{K}^n \\ \|y\|_2 \leq 1, \ B^*y=0}} \text{Re}\langle y, w \rangle. \tag{6}$$

In particular, the following are equivalent:

(i) $Bc = w$ for some $c \in \mathbb{K}^m$;

(ii) $r_* = 0$;

(iii) $\langle y, w \rangle = 0$ for every $y \in \ker B^*$.

If $r_* \neq 0$, then

$$y_* = \frac{r_*}{\|r_*\|_2}$$

is a unit dual obstruction satisfying

$$B^* y_* = 0, \quad \operatorname{Re} \langle y_*, w \rangle = \|r_*\|_2.$$

Proof. The orthogonal decomposition

$$w - Bc = r_* + (q_* - Bc), \quad r_* \perp \operatorname{ran} B,$$

gives

$$\|w - Bc\|_2^2 = \|r_*\|_2^2 + \|q_* - Bc\|_2^2.$$

This proves (5); equality is attained, for example, at $c = B^\dagger w$.

If $B^* y = 0$ and $\|y\|_2 \leq 1$, then

$$\operatorname{Re} \langle y, w \rangle = \operatorname{Re} \langle y, r_* \rangle \leq \|r_*\|_2.$$

When $r_* \neq 0$, equality holds at y_* . When $r_* = 0$, the maximum is zero because $y = 0$ is feasible. This proves (6) and the equivalences. $\square$

Corollary 3.2 (A posteriori lower bound). *Any vector y with $B^* y = 0$ and $\|y\|_2 \leq 1$ certifies*

$$\|w - Bc\|_2 \geq \operatorname{Re} \langle y, w \rangle$$

for every c . Thus a rational or interval-verified y can certify a nonzero cover defect without computing a pseudoinverse.

3.2 Nonnegative coefficients

Suppose now that B and w are real and that $\mathcal{C} = \mathbb{R}_+^m$. The attainable set

$$K = B\mathbb{R}_+^m$$

is a finitely generated closed convex cone.

Theorem 3.3 (Farkas and quantitative cone certificates). *Exactly one of the following alternatives holds:*

(a) *there is $c \geq 0$ such that $Bc = w$;*

(b) *there is $y \in \mathbb{R}^n$ such that $B^\top y \geq 0$ and $\langle y, w \rangle < 0$.*

Moreover,

$$\operatorname{dist}(w, K) = \max_{\substack{y \in \mathbb{R}^n \\ \|y\|_2 \leq 1, \ B^\top y \leq 0}} \langle y, w \rangle. \quad (7)$$

If $q = P_K w$, $r = w - q$, and $r \neq 0$, then

$$B^\top r \leq 0, \quad \langle r, q \rangle = 0,$$

so $-r$ is a Farkas obstruction and $r/\|r\|_2$ attains (7).

Proof. The alternative is the finite-dimensional Farkas lemma [4]. For the quantitative statement, the projection inequality for the closed convex cone K gives

$$\langle w - q, z - q \rangle \leq 0 \quad (z \in K).$$

Taking $z = 0$ and $z = 2q$ yields $\langle r, q \rangle = 0$. Taking $z = tBe_j$, $t \geq 0$, gives $B^\top r \leq 0$. Hence the right side of (7) is at least $\|r\|_2$. Conversely, if $B^\top y \leq 0$, then $\langle y, z \rangle \leq 0$ for all $z \in K$, and therefore

$$\langle y, w \rangle \leq \langle y, w - q \rangle \leq \|r\|_2$$

for $\|y\|_2 \leq 1$. This proves the formula. Finally,

$$\langle -r, w \rangle = -\langle r, q + r \rangle = -\|r\|_2^2 < 0,$$

so $-r$ has the required Farkas signs. $\square$

3.3 General convex rules with closed attainable images

For a nonempty set $\mathcal{C} \subseteq \mathbb{K}^m$, define its support function

$$\sigma_{\mathcal{C}}(u) = \sup_{c \in \mathcal{C}} \operatorname{Re} \langle u, c \rangle.$$

The projection and support-function formulation below is standard finite-dimensional convex analysis [1, 3].

Theorem 3.4 (Convex allowed-set dual). *Let $\mathcal{C}$ be nonempty and convex, and assume $B\mathcal{C}$ is closed. Then*

$$\operatorname{dist}(w, B\mathcal{C}) = \max_{\|y\|_2 \leq 1} \{\operatorname{Re} \langle y, w \rangle - \sigma_{\mathcal{C}}(B^*y)\}. \quad (8)$$

In particular, an allowed exact cover exists if and only if the right side is zero.

Proof. Put $K = B\mathcal{C}$, and let $q = P_K w$. For every $\|y\|_2 \leq 1$ and $z \in K$,

$$\operatorname{Re} \langle y, w \rangle - \sigma_K(y) \leq \operatorname{Re} \langle y, w - z \rangle \leq \|w - z\|_2.$$

Taking the infimum in z gives the upper bound by $\operatorname{dist}(w, K)$. If $w \notin K$, the projection inequality shows that $y = (w - q)/\|w - q\|_2$ supports K at q ; hence $\sigma_K(y) = \operatorname{Re} \langle y, q \rangle$, and equality holds. If $w \in K$, $y = 0$ gives equality at zero. Finally, $\sigma_K(y) = \sigma_{\mathcal{C}}(B^*y)$. $\square$

Remark 3.5 (Closedness). The image of an arbitrary closed convex set under a linear map need not be closed. The hypothesis in [theorem 3.4](#) is automatic for the coefficient sets used most often here: compact polytopes and polyhedra. It is stated explicitly rather than hidden in the projection notation.

4 Bounded overlap does not imply coverage

Bounded overlap provides an upper-norm inequality. It does not provide a lower frame bound, surjectivity, or coefficient feasibility.

Lemma 4.1 (What overlap actually controls). *Put $S_j = \{\omega \in \Omega : I_{\omega j} = 1\}$. For the weighted incidence matrix (1),*

$$\|Bc\|_2^2 \leq \Delta(I) \sum_{\omega \in \Omega} \sum_{j=1}^m I_{\omega j} |g_{\omega j} c_j|^2. \quad (9)$$

In particular, if $|g_{\omega j}| \leq G$, then

$$\|Bc\|_2^2 \leq \Delta(I) G^2 \sum_{j=1}^m |c_j|^2 |S_j|.$$

Proof. For each row ω , Cauchy–Schwarz gives

$$\left| \sum_j I_{\omega j} g_{\omega j} c_j \right|^2 \leq \left(\sum_j I_{\omega j} \right) \sum_j I_{\omega j} |g_{\omega j} c_j|^2.$$

Sum over ω . The last displayed specialization follows by interchanging the sums and using $\sum_{\omega} I_{\omega j} = |S_j|$. $\square$

Remark 4.2. The right side of (9) is an upper synthesis bound. No reverse inequality follows from $\Delta(I) < \infty$: columns may be linearly dependent, miss directions, or conflict with the target weights.

4.1 A disjoint full-support family with fixed residual

Theorem 4.3 (Overlap one, but no asymptotic cover). *For every $n \geq 1$, there is an unweighted incidence matrix $B_n \in \{0, 1\}^{2n \times n}$ and a positive target $w_n \in \mathbb{R}^{2n}$ such that:*

- (i) *every atom lies in exactly one block, so $\Delta(I) = 1$ and the blocks partition the full support;*
- (ii) *no signed, complex, or nonnegative exact cover exists;*
- (iii) *the exact least-squares defect is*

$$\text{dist}(w_n, \text{ran } B_n) = \sqrt{\frac{n}{2}};$$

- (iv) *the relative defect is the fixed constant*

$$\frac{\text{dist}(w_n, \text{ran } B_n)}{\|w_n\|_2} = \frac{1}{\sqrt{10}}.$$

Proof. Let

$$b_j = e_{2j-1} + e_{2j}, \quad w_n = \sum_{j=1}^n (e_{2j-1} + 2e_{2j}).$$

The columns have disjoint supports and every row is present once. For any $c = (c_j)$,

$$\|w_n - B_n c\|_2^2 = \sum_{j=1}^n (|1 - c_j|^2 + |2 - c_j|^2).$$

Each summand is uniquely minimized at $c_j = 3/2$, where it equals $1/2$. The minimizer is nonnegative, so imposing $c \geq 0$ does not improve or worsen the minimum. Hence the squared defect is $n/2$, whereas $\|w_n\|_2^2 = 5n$. The positive defect excludes every exact cover. $\square$

Corollary 4.4 (No approximate-cover consequence). *Even the conjunction of bounded overlap, full set coverage, positive target weights, and a disjoint block partition does not imply*

$$\text{dist}(w_X, \text{ran } B_X) = o(\|w_X\|_2).$$

4.2 Signed feasibility does not imply allowed feasibility

Theorem 4.5 (A sharp two-column cone obstruction). *Let*

$$B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad w = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Then:

- (i) *every row occurs in a block and the overlap degree is two;*

- (ii) $Bc = w$ has the unique signed solution $c = (1, -1)^\top$;
- (iii) no nonnegative cover exists;
- (iv)

$$\text{dist}(w, B\mathbb{R}_+^2) = \frac{1}{\sqrt{2}}.$$

The vector $y = (-1, 1)^\top$ is a Farkas certificate:

$$B^\top y = (0, 1)^\top \geq 0, \quad \langle y, w \rangle = -1 < 0.$$

Proof. The signed equation gives $c_1 = 1$ from the first row and $c_1 + c_2 = 0$ from the second. For $c_1, c_2 \geq 0$,

$$\|w - Bc\|_2^2 = (1 - c_1)^2 + (c_1 + c_2)^2.$$

At fixed c_1 , the minimum in $c_2 \geq 0$ occurs at $c_2 = 0$. The remaining quadratic is minimized at $c_1 = 1/2$, with value $1/2$. The displayed Farkas signs are immediate. $\square$

4.3 Native-key collapse can create a false cover

Let $A : \mathbb{K}^\Omega \rightarrow \mathbb{K}^\Gamma$ aggregate literal atoms into coarse classes, for example by summing all atoms with equal numerical determinant.

Proposition 4.6 (Coarse feasibility need not lift). *The coarse equation*

$$ABc = Aw \tag{10}$$

is necessary but not sufficient for the literal equation $Bc = w$. The failure can occur with one block, one coarse key, and overlap one.

Proof. Take two literal atoms with the same coarse key,

$$A = \begin{pmatrix} 1 & 1 \end{pmatrix}, \quad B = e_1, \quad w = e_2.$$

Then $AB = Aw = 1$, so $c = 1$ solves the coarse equation. But $Bc = e_1 \neq e_2 = w$, and the literal defect is $\|e_2 - e_1\|_2 = \sqrt{2}$. The two atoms may have the same determinant and different provenance; aggregation has hidden exactly that difference. $\square$

Remark 4.7 (Union-frame rule). If a candidate column emits atoms outside $\text{supp } w$, the correct universe is the union of target and column supports, with target coefficient zero on the extra rows. Deleting those rows can turn leakage into a false exact cover, just as [proposition 4.6](#) turns a literal mismatch into a coarse identity.

Example 4.8 (Overlap two and inconsistent multiplicity). *Let*

$$B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad w = (1, 1, 1)^\top.$$

The first and third rows force $c_1 = c_2 = 1$, while the middle row would then equal 2. Thus there is no exact cover although every atom is covered and the overlap is two. The least-squares solution is $c_1 = c_2 = 2/3$, with residual $(1, -1, 1)^\top/3$ and relative defect $1/3$. It gives a three-row binary example of why an ad hoc division by overlap multiplicity cannot replace the literal system.

5 Least-squares stability and exact normalization

5.1 Conditioning of the unrestricted certificate

Assume $B \neq 0$, and let $\sigma_+(B)$ be its smallest positive singular value.

Proposition 5.1 (Minimum-norm synthesis and target stability). *The vector*

$$c^\dagger = B^\dagger w$$

is the unique minimum-norm least-squares coefficient, and

$$\|c^\dagger\|_2 \leq \frac{\|P_B w\|_2}{\sigma_+(B)} \leq \frac{\|w\|_2}{\sigma_+(B)}. \quad (11)$$

For two targets w, w' ,

$$|\text{dist}(w, \text{ran } B) - \text{dist}(w', \text{ran } B)| \leq \|w - w'\|_2, \quad (12)$$

$$\|B^\dagger w - B^\dagger w'\|_2 \leq \frac{\|w - w'\|_2}{\sigma_+(B)}. \quad (13)$$

Proof. The singular-value decomposition gives the minimum-norm property and $\|B^\dagger\|_{2 \rightarrow 2} = \sigma_+(B)^{-1}$, proving [equations \(11\) and \(13\)](#). The distance to a fixed nonempty closed set is 1-Lipschitz:

$$\text{dist}(w, \text{ran } B) \leq \|w - w'\|_2 + \text{dist}(w', \text{ran } B),$$

and the reverse inequality follows by interchanging w, w' . $\square$

Remark 5.2 (Residual stability versus coefficient stability). The residual defect is always stable with constant one. The synthesis coefficients may be unstable when $\sigma_+(B)$ is small. Therefore a numerically tiny residual is not by itself a well-conditioned symbolic certificate.

Proposition 5.3 (Column perturbation with bounded coefficients). *Let $\tilde{B} = B + E$. If $Bc = w$, then*

$$\text{dist}(w, \text{ran } \tilde{B}) \leq \|E\|_{2 \rightarrow 2} \|c\|_2. \quad (14)$$

More generally, if $\emptyset \neq \mathcal{C}$ is contained in the radius- R ball, then

$$\left| \text{dist}(w, B\mathcal{C}) - \text{dist}(w, \tilde{B}\mathcal{C}) \right| \leq R\|E\|_{2 \rightarrow 2}. \quad (15)$$

No uniform bound of this form follows for an unbounded coefficient cone without an a priori bound on the relevant coefficients.

Proof. Using the same coefficient c ,

$$\|w - \tilde{B}c\|_2 = \|Ec\|_2,$$

which proves [\(14\)](#). For every $c \in \mathcal{C}$,

$$\|Bc - \tilde{B}c\|_2 \leq R\|E\|_{2 \rightarrow 2}.$$

Thus the two attainable sets have Hausdorff distance at most the right side, and their distances from w differ by no more than that amount. $\square$

5.2 The exact normalization identity

Let $\mathcal{N}_X > 0$ be the original global normalizer and let $\Phi_{\omega, X} \in \mathbb{C}$ be the physical contribution of atom ω . Define the normalized linear statistic

$$\mathcal{T}_X(v) = \frac{1}{\mathcal{N}_X} \sum_{\omega \in \Omega_X} v_\omega \Phi_{\omega, X}. \quad (16)$$

No atom-dependent renormalization is allowed in this definition.

Theorem 5.4 (Exact reassembly and coefficient cost). *If $Bc = w$ exactly, then*

$$\mathcal{T}_X(w) = \sum_{j=1}^m c_j \mathcal{T}_X(b_j). \quad (17)$$

If, on the same normalization,

$$\mathcal{T}_X(b_j) = M_j + O(E_j),$$

then

$$\mathcal{T}_X(w) = \sum_{j=1}^m c_j M_j + O\left(\sum_{j=1}^m |c_j| E_j\right). \quad (18)$$

In particular, a uniform block error E is amplified by at most $\|c\|_1 E$.

Proof. Apply the same linear functional $\mathcal{T}_X$ to $Bc = w$. The error estimate follows from the triangle inequality. $\square$

Definition 5.5 (Exact-cover amplification). For an allowed exact cover, define

$$\tau_{\text{cov}}(B, w; \mathcal{C}) = \inf_{\substack{c \in \mathcal{C} \\ Bc = w}} \|c\|_1,$$

with value $+\infty$ when no exact cover exists. When block errors are nonuniform, the intrinsic cost is instead

$$\inf_{\substack{c \in \mathcal{C} \\ Bc = w}} \sum_j |c_j| E_j.$$

Remark 5.6 (Logarithmic estimates). If every block has error $O(X(\log X)^{-A})$ for arbitrary fixed A , then a cover cost $\|c_X\|_1 = O((\log X)^C)$ preserves arbitrary logarithmic saving after increasing A . The weaker statement $\|c_X\|_1 = X^{o(1)}$ is not enough for that conclusion: $\exp(\sqrt{\log X}) = X^{o(1)}$ dominates every fixed power of $\log X$. The exact product $\|c_X\|_1 E_X$ must be checked.

5.3 Why an approximate cover is a different theorem

Proposition 5.7 (Residual remainder identity). *For an arbitrary coefficient c , put $r = w - Bc$. Then*

$$\mathcal{T}_X(w) = \sum_j c_j \mathcal{T}_X(b_j) + \mathcal{T}_X(r). \quad (19)$$

The Euclidean statement $\|r\|_2 = o(\|w\|_2)$ does not, by itself, imply $\mathcal{T}_X(r) = o(1)$ or any prescribed physical error bound.

Proof. The identity follows by linearity. For the obstruction, let $\varepsilon_n \downarrow 0$, take one block $B_n = e_2$, $c_n = 1$, and

$$w_n = e_2 + \varepsilon_n e_1, \quad r_n = w_n - B_n c_n = \varepsilon_n e_1.$$

Define $\mathcal{T}_n(v) = \varepsilon_n^{-1} v_1$. Then $\|r_n\|_2 / \|w_n\|_2 \rightarrow 0$, but $\mathcal{T}_n(r_n) = 1$. Equivalently, the only automatic estimate is

$$|\mathcal{T}_X(r)| \leq \|\mathcal{T}_X\|_{2 \rightarrow \mathbb{C}} \|r\|_2,$$

and the physical operator norm has not been bounded by an incidence calculation. $\square$

Corollary 5.8 (Exactness threshold for a pure cover argument). *A block-cover argument transports the original physical statistic without a new remainder estimate only when $w \in B\mathcal{C}$ in the literal frame. When $B\mathcal{C}$ is closed, this is equivalent to $\varepsilon_{\text{cov}} = 0$. If no exact allowed cover is used, closure requires an independent bound for $\mathcal{T}_X(r)$ on the explicitly identified residual.*

Remark 5.9. This corollary does not say that approximate covers are useless. It says that they are not exact covers. A physical norm estimate for the residual could be valuable, but it is an additional arithmetic input and must be labeled as such.

6 An exact sufficient certificate for the TPC-18 cover gate

TPC-18 proves a stopping dichotomy inside an admissible symmetric tail block and states a conditional global implication under two additional inputs [5]: a primitive generic determinant-dispersion estimate and an all-block representation of the upstream hard packet up to $o(X)$ in the physical statistic. The present section gives an exact zero-remainder certificate for the second input and records the distinct controlled-residual route.

6.1 The physical incidence system

For a fixed nonzero h_0 , let w_X be the literal atomic coefficient of the hard packet inherited from the upstream normal form. Construct Ω_X from the union of its active atoms and the atoms of every proposed TPC-18 block. Let $b_{j,X}$ be the literal coefficient of the j -th candidate block, including both the prefix and complementary-tail pieces, and form

$$B_X = (b_{1,X} \ \cdots \ b_{m_X,X}).$$

The upstream derivation also determines an allowed coefficient set $\mathcal{C}_X$. It may be a singleton, a partition-of-unity polytope, or another explicitly derived set.

Definition 6.1 (Exact TPC-18 full-block certificate). An exact full-block cover certificate at scale X consists of:

- (i) the complete literal row list Ω_X , with native keys and zero target coefficients retained on every leakage row;
- (ii) the exact weighted incidence matrix B_X ;
- (iii) an allowed coefficient $c_X \in \mathcal{C}_X$ satisfying

$$B_X c_X = w_X;$$

- (iv) a proof-producing exact verification of the equality, together with a rigorous bound for the coefficient amplification in the physical normalization.

Remark 6.2 (The original unit cover). If the upstream normal form requires the unweighted sum of all candidate blocks, then $\mathcal{C}_X = \{\mathbf{1}\}$, and the certificate is the single identity

$$B_X \mathbf{1} = w_X.$$

Finding some $c_X \in \mathbb{R}^{m_X}$ is not a substitute. Allowing different coefficients changes the decomposition unless that freedom has been derived upstream. Failure of the exact identity $B_X \mathbf{1} = w_X$ does not by itself refute the original TPC-18 hypothesis: the separate controlled-residual route may still prove $\mathcal{T}_X(w_X - B_X \mathbf{1}) = o(X)$.

Theorem 6.3 (Conditional exact-cover transfer). *Assume an exact TPC-18 full-block certificate in the sense of [definition 6.1](#). Let the normalized physical functional be $\mathcal{T}_X$, and suppose each certified block has a decomposition*

$$\mathcal{T}_X(b_{j,X}) = M_{j,X} + E_{j,X}.$$

Then

$$\mathcal{T}_X(w_X) = \sum_{j=1}^{m_X} c_{j,X} M_{j,X} + O\left(\sum_{j=1}^{m_X} |c_{j,X}| |E_{j,X}|\right). \quad (20)$$

Consequently, if the upstream calibrated main term equals $\sum_j c_{j,X} M_{j,X}$ and the displayed weighted error is admissible, the all-block reassembly step is valid with no overlap correction beyond $B_X c_X = w_X$.

Proof. This is [theorem 5.4](#) applied on the literal TPC-18 frame. The hypotheses about the calibrated main term and error are stated separately because they are not consequences of incidence feasibility. $\square$

Proposition 6.4 (Controlled-residual route). *Let c_X be the coefficient prescribed by the upstream decomposition, put $r_X = w_X - B_X c_X$, and retain the original physical functional $\mathcal{T}_X$. Then*

$$\mathcal{T}_X(w_X) = \sum_j c_{j,X} \mathcal{T}_X(b_{j,X}) + \mathcal{T}_X(r_X).$$

Thus a theorem proving $\mathcal{T}_X(r_X) = o(X)$, together with the required block estimates and the calibrated main term, can verify the original asymptotic all-block hypothesis even when $r_X \neq 0$. Euclidean smallness of r_X alone does not supply this theorem.

Proof. The identity is [proposition 5.7](#). The remaining statements distinguish an assumed physical remainder estimate from the Euclidean incidence calculation. $\square$

Corollary 6.5 (A checkable failure certificate). *If unrestricted synthesis is permitted and*

$$r_X = (I - B_X B_X^\dagger) w_X \neq 0,$$

then $r_X / \|r_X\|_2$ certifies that the proposed block family cannot exactly cover the TPC-18 hard packet. If nonnegative synthesis is required, any Farkas vector from [theorem 3.3](#) certifies failure of that allowed exact cover. Such a certificate is a rigorous negative L1 result for the zero-remainder route; it does not rule out a controlled physical residual and is not an arithmetic cancellation theorem.

6.2 What this does not supply

Proposition 6.6 (Independent downstream inputs). *Even an exact, well-conditioned full-block certificate does not imply:*

- (i) *the TPC-18 primitive generic determinant-dispersion estimate;*
- (ii) *a determinant-energy lower bound or distinguished zero-mode cancellation estimate for a later post-bin coefficient;*
- (iii) *the prefix estimate for a block outside the published TPC-17 geometry;*
- (iv) *an admissible bound for an approximate-cover residual;*
- (v) *a fixed-shift Hardy–Littlewood asymptotic.*

Proof. The certificate is the coefficient identity $B_X c_X = w_X$. Items (i)–(iii) are analytic estimates on the columns, item (iv) is an estimate on a vector outside their exact synthesis, and item (v) requires the full upstream and downstream theorem chain. None is a linear-algebraic consequence of column-space membership. $\square$

Remark 6.7 (Present status). This paper does not enumerate the full physical atom list of the TPC-16/TPC-18 normal form, exhibit an allowed exact c_X , or bound the physical image of a nonzero residual. Therefore the TPC-18 all-block representation remains an independent open gate. The result here is a rigorous format in which a future exact certificate, controlled-residual theorem, or negative exact-route certificate can be stated and checked.

Check	Exact question	Status in this paper
Support overlap	Does every target atom occur, and how many columns touch it?	Necessary diagnostic only
Literal cover	Does an allowed c_X satisfy $B_X c_X = w_X$ on all native rows?	Exact certificate schema; not instantiated globally
Coefficient cost	Is $\sum_j c_{j,X} E_{j,X} $ within the physical error?	Exact accounting formula
Primitive determinant dispersion	Are the TPC-18 generic residual columns small at fixed h_0 ?	Independent unproved arithmetic input
Full theorem chain	Do all upstream identities, block estimates, and endpoint budgets hold simultaneously?	Not proved

Table 1: Set coverage, coefficient coverage, and arithmetic cancellation are different gates.

7 Separate feasibility, determinant, and physical ledgers

Several losses can occur near the same endpoint, but they measure different objects. We record them separately to prevent an incidence gain from being counted as arithmetic cancellation.

7.1 The cover ledger

The exact feasibility entry is binary:

$$\boxed{w_X \in B_X \mathcal{C}_X.} \quad (21)$$

When $B_X \mathcal{C}_X$ is closed, this is equivalent to $\varepsilon_{\text{cov}}(B_X, w_X; \mathcal{C}_X) = 0$. The original asymptotic TPC-18 representation also admits a distinct route in which the prescribed coefficient leaves $r_X \neq 0$ but a new theorem proves $\mathcal{T}_X(r_X) = o(X)$. When it holds, the quantitative reassembly cost is the exact product

$$\sum_j |c_{j,X}| |E_{j,X}|,$$

minimized over allowed exact covers if there is a choice. If $\|c_X\|_1 = X^{\lambda_{\text{cov}}+o(1)}$ with $\lambda_{\text{cov}} > 0$, that amplification is a physical-interface loss only when it is not already present in the block error. It is never a zero-mode saving.

7.2 The determinant/zero-mode ledger

On a separate conditional post-bin branch, suppose

$$S_X = X^{o(1)} Q_X, \quad |Z_X| \leq X^{-\eta_Z+o(1)} Q_X^2, \quad D_X \geq X^{-\lambda_D+o(1)} Q_X^3 J_X^2.$$

These are hypotheses about a literal coefficient, not conclusions of the incidence model. Its principal angle

$$\rho_X = \frac{|Z_X|}{\sqrt{S_X D_X}}$$

then satisfies

$$\rho_X \leq \frac{X^{\lambda_D/2-\eta_Z+o(1)}}{J_X}.$$

Thus the recorded estimates certify the critical angle scale under

$$\boxed{\lambda_D \leq 2\eta_Z.} \quad (22)$$

This is the inherited determinant/zero-mode compatibility rule [6]. Solving $B_X c_X = w_X$ neither improves λ_D nor proves $\eta_Z > 0$.

7.3 The strict physical-loss ledger

Let Λ_{phys} be the sum of fixed-power losses actually incurred by the full physical reduction and downstream comparison, excluding every loss already represented in $\lambda_D - 2\eta_Z$. The route remains viable only under the strict rule

$$\boxed{\Lambda_{\text{phys}} < \frac{1}{400}}. \quad (23)$$

Equality or excess is a stop condition. If a polynomial exact-cover amplification is newly incurred, it is charged here once. It may not also be inserted into λ_D or subtracted from η_Z .

Theorem 7.1 (No automatic conversion among ledgers). *None of the following implications follows from the results of this paper:*

- (i) $w_X \in B_X \mathcal{C}_X \implies \lambda_D \leq 2\eta_Z$;
- (ii) $\lambda_D \leq 2\eta_Z \implies w_X \in B_X \mathcal{C}_X$;
- (iii) *bounded overlap* $\implies \Lambda_{\text{phys}} = 0$;
- (iv) *a small Euclidean cover residual* $\implies$ *a saving in η_Z* ;
- (v) *determinant-compatible column metadata* $\implies$ *the TPC-18 generic determinant estimate*.

Proof. The cover ledger concerns membership of one target vector in an allowed linear image. The determinant ledger compares a zero mode with an energy of a literal post-bin coefficient. The physical ledger accumulates losses in arithmetic reassembly, and the TPC-18 generic estimate bounds a fixed-shift correlation inside each compatible column. The counterexamples in [section 4](#) already separate support overlap from linear feasibility, while [proposition 5.7](#) separates Euclidean residual size from a physical statistic. No theorem identifies the remaining quantities, so no listed conversion follows. $\square$

Ledger	Required statement	Status here
Literal zero-remainder cover	Exact membership $w_X \in B_X \mathcal{C}_X$ on all native rows	Sufficient zero-remainder route; physical feasibility unproved
Coefficient amplification	Weighted block error fits the target error	Exact formula; no physical c_X constructed
Determinant/zero mode	$\lambda_D \leq 2\eta_Z$	Independent conditional post-bin ledger
Physical endpoint	$\Lambda_{\text{phys}} < 1/400$, strictly	Independent inherited stop rule
TPC-18 arithmetic	Primitive generic determinant dispersion at the same fixed h_0	Independent unproved L2 input

Table 2: The same gain or loss may not be moved between rows without a literal crosswalk theorem.

Remark 7.2 (TPC-18 gate independence). The all-block representation in the first row is still its own gate. The present paper makes the exact zero-remainder route falsifiable by an exact residual or Farkas vector, but does not declare the original asymptotic gate closed: a separate controlled-residual route may remain. Even if the representation gate were closed, the last row would remain open.

8 Audited conclusion and next certificate

Theorem 8.1 (Audited finite-dimensional conclusion). *Let w be a literal finite atom coefficient, let B contain the exact candidate block columns, and let $\mathcal{C}$ encode the allowed synthesis coefficients.*

(i) *For unrestricted coefficients,*

$$\varepsilon_{\text{cov}} = \|(I - BB^\dagger)w\|_2,$$

and, when the residual is nonzero, its normalized direction is an optimal dual obstruction.

(ii) *For nonnegative real coefficients, Farkas' alternative is exact and*

$$\varepsilon_{\text{cov}} = \max_{\substack{\|y\|_2 \leq 1 \\ B^\top y \leq 0}} \langle y, w \rangle.$$

(iii) *For a closed convex attainable image, the support-function formula (8) is exact.*

(iv) *Bounded overlap, even overlap one with every atom covered, does not imply an exact or $o(1)$ -relative cover.*

(v) *An exact cover transports the original global normalization with coefficient cost $\sum_j |c_j| E_j$. An approximate cover leaves the independent physical remainder $\mathcal{T}_X(w - Bc)$.*

Proof. Combine [theorems 3.1, 3.3, 3.4, 4.3](#) and [5.4](#) and [proposition 5.7](#). □

What is proved. The literal synthesis model, exact residual formulas, dual certificates, explicit finite counterexamples, target and column stability bounds, and coefficient-amplification identity are unconditional L0 results. They provide an exact L1 format for auditing a proposed TPC-18 cover.

What is not proved. No complete physical atom list w_X , incidence matrix B_X , or allowed exact solution c_X is constructed for the full TPC-18 normal form, and no nonzero residual is controlled in the physical statistic. No primitive generic determinant-dispersion estimate, prefix extension, determinant-energy survival theorem, or distinguished zero-mode estimate is proved. Hence the TPC-18 all-block representation and every L2 arithmetic gate remain open.

Support and key rule. The certificate must be checked on the union of actual target and candidate-block supports. Native atoms may not be merged merely because they have the same determinant. Every extra column atom is a row with target coefficient zero. A certificate after aggregation is only necessary unless a separate lifting theorem is proved.

Coefficient rule. The allowed set $\mathcal{C}_X$ comes from the upstream decomposition. A signed solution does not certify a nonnegative cover, and an arbitrary solution does not certify the original unit all-block identity. Exact feasibility and coefficient amplification must both be verified.

Normalization rule. Only $B_X c_X = w_X$ gives pure algebraic transport under the original global normalizer. A least-squares residual, however small in ℓ^2 , requires a new physical remainder theorem. Such a theorem could verify the original asymptotic TPC-18 route, but the residual may not be discarded by bounded overlap or numerical conditioning.

Ledger rule. The exact-cover condition $w_X \in B_X \mathcal{C}_X$ (equivalently $\varepsilon_{\text{cov}} = 0$ when the attainable image is closed), the post-bin determinant rule $\lambda_D \leq 2\eta_Z$, and the strict endpoint rule $\Lambda_{\text{phys}} < 1/400$ are independent. Equality or excess in the last rule stops the route. No gain or loss is charged twice.

Arithmetic claim boundary. All statements retain one prescribed nonzero h_0 . Nothing specializes it to 2, proves a fixed-shift Hardy–Littlewood asymptotic, gives a prime-pair lower bound, breaches the parity barrier, or implies the twin-prime conjecture.

Next exact task. A genuine advance on this gate must export the upstream hard packet and all candidate TPC-18 blocks to the same literal row list, then do one of the following:

- (a) produce an exact symbolic or interval-verified allowed coefficient c_X with $B_X c_X = w_X$ and acceptable amplification;
- (b) produce a dual residual or Farkas vector proving that the proposed block family cannot exactly cover the packet; or
- (c) isolate an explicit residual r_X and prove a new physical bound for $\mathcal{T}_X(r_X)$.

The first would close only the exact zero-remainder interface. The second would rule out that block family only as an exact zero-remainder route; it would not exclude a separately controlled residual route. The third would define such a residual route. None, by itself, proves the independent TPC-18 determinant-dispersion input.

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